

Result

$$\frac{\Delta\nu}{\nu} \approx 2.46 \times 10^{-15}$$

Confirmed to 0.1%

Blueshift

$$\nu_{\text{bot}} > \nu_{\text{top}}$$
$$\Delta\nu/\nu = gH/c^2$$

$$H = 22.5 \text{ m}$$

Source (^{57}Fe)

$$P_{\text{top}} > P_{\text{bot}}$$

Higher pressure $\Rightarrow$ faster clock

γ photon $\downarrow$

Absorber (^{57}Fe)

$$P_{\text{bot}} < P_{\text{top}}$$

∇P
IS pressure gradient

Earth surface